

## SEMESTER END / BACKLOG SUBJECT EXAMINATIONS - JANUARY 2026

|             |                                                     |            |                |
|-------------|-----------------------------------------------------|------------|----------------|
| Program     | : <b>B.E. - Information Science and Engineering</b> | Semester   | : <b>V</b>     |
| Course Name | : <b>Software Engineering</b>                       | Max. Marks | : <b>100</b>   |
| Course Code | : <b>21 / 22 / 23IS51</b>                           | Duration   | : <b>3 Hrs</b> |

### Instructions to the Candidates:

- Answer one full question from each unit.

### UNIT - I

- With a neat diagram explain spiral model and its advantages and disadvantages over waterfall model briefly. CO1 (08)
  - Describe the key principles of professional software development. CO1 (06)
  - List out different ethical responsibilities of software engineering profession. CO1 (06)
- State and explain four different phases of Rational Unified Process (RUP) with a neat diagram. CO1 (08)
  - Demonstrate how plan-driven and agile approaches can be combined in a hybrid software development process for a real-time system. CO1 (06)
  - What is the need of extreme programming? Draw and describe the extreme programming cycle. CO1 (06)

### UNIT - II

- Clearly distinguish between different categories of requirements with examples. CO2 (06)
  - Outline the main sections or contents typically included in the software requirements document. CO2 (08)
  - Explain the different ways to collect requirements in software engineering. CO2 (06)
- Explain the Requirements engineering process. CO2 (10)
  - Consider an automated traffic signal system. Write functional and non-functional requirements for this system. CO2 (10)

### UNIT - III

- Draw the Use case diagram, class diagram and sequence diagram for Library Management System. CO3 (08)
  - Consider a software company, "CollabSoft," wants to build a web-based application similar to Google Docs, allowing multiple users to edit the same document simultaneously in real-time. Which architectural pattern (e.g., MVC, Client-Server, Peer-to-Peer) is most suitable for the core real-time functionality? Justify your choice. CO3 (07)
  - Compare and contrast the Repository and Client-Server architectural patterns. Illustrate each with a suitable diagram CO3 (05)

6. a) Discuss the software architecture of an ATM system. Illustrate the functions of the input, process and output components. CO3 (08)  
b) What is the importance of design pattern? With the diagram explain the UML model of Observer design Pattern. CO3 (07)  
c) Discuss open source software development with an example. CO3 (05)

## UNIT- IV

7. a) Demonstrate equivalence class test cases for the NextDate function. CO4 (08)  
b) Explain the different levels of Boundary Value Testing (BVT) with a suitable example. CO4 (07)  
c) List any four common categories from a fault taxonomy. Provide one example of a potential fault in a simple 'User Login' module. CO4 (05)
8. a) For a function that determines the type of a triangle given three integer sides (a, b, c), design Equivalence Class test cases. Clearly identify the valid and invalid equivalence classes CO4 (08)  
b) Discuss the various 'Levels of Testing' in the software development lifecycle. Briefly explain the objective and the typical scope (what is tested) for any three levels. CO4 (07)  
c) Compare and contrast Specification-Based (Black-Box) and Code-Based (White-Box) testing strategies. CO4 (05)

## UNIT - V

9. a) Explain the key activities involved in project risk management. How do risk identification, analysis, and mitigation contribute to project success? CO5 (08)  
b) Explain plan-driven development. What are the advantages and limitations of using a plan-driven approach in software projects? CO5 (06)  
c) Describe the role of software measurement and metrics in quality management. CO5 (06)
10. a) Apply principles of people management to handle conflict within a development team working under tight deadlines. Provide a suitable action plan. CO5 (06)  
b) Choose a suitable software pricing strategy (cost-based, market-based, value-based) for developing a custom ERP system for a small business and justify your choice. CO5 (06)  
c) List out three categories of risks with an example and explain the stages of risk management with diagram. CO5 (08)

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